

CeVRicale: A VR app for Cervical Rehabilitation

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ABSTRACT

We propose CeVRicale, a cervical rehabilitation application based on the use of virtual reality (VR). CeVRicale is smartphone-based, thus it may be available to larger shares of population when compared to those applications that are implemented for head mounted displays such as HTC Vive or Oculus Rift. The app exploits a smartphone's sensor to track head movements in five exergames inspired by rehabilitation exercises. This project is the first step in a study to evaluate the effectiveness and efficiency of a low cost VR application in the treatment of cervical musculoskeletal disorders.

Index Terms: Health informatics—Health care information systems—Interactive games—Virtual Reality

1 INTRODUCTION AND BACKGROUND

Nowadays, the amount of physical activity (PA) in which a healthy subject engages is almost always below the World Health Organization's recommended standard [6]. The video gaming world may offer an important contribution to invert such a trend with the development of serious games and exergames, i.e. exercises in the form of a game. The use of Virtual Reality (VR) in exergames has shown to enhance their attractiveness and engagement [3, 7]. However, not all exergames can be played by everyone, especially in the case of patients whose pathology requires rehabilitation; nonetheless, VR has been used for long in the rehabilitation field, because it allows a personalized treatment, less physiotherapist supervision and at-home exercise. These characteristics appear particularly desirable during the SARS-COV2 pandemic, which imposed on health operators the enforcement of at-distance activities, e.g. tele-health or at-home supervised treatment [1]. The use of VR in home-based rehabilitation of musculoskeletal disorders is still limited by economic problems. Indeed, public health systems do not have enough resources to provide suitable technological instruments to many patients. The development of smartphone applications for tele-health could significantly reduce costs and therefore allow for a vast diffusion. Keeping this goal in mind, we developed an application for a head-mounted device (HMD) that uses VR-based exergames to guide a patient to perform some complementary exercises to the rehabilitation treatment for neck musculoskeletal disorders.

2 METHOD

We researched exergame apps for cervical area which could be adapted to a subject during rehabilitation. Then, we designed five games based on relevant literature to include all movement and functions of the cervical spine [4, 8]. The cervical spine supports movement along 3 axes with 3 degrees of freedom:

- flexion-extension (tilting the head towards the shoulder);

- inclination;
- rotation.

These movements are the result of the interaction between upper and lower cervical spine. Each game is equipped with two metrics, to assess the achievements of a patient during the treatment. Due to intrinsic limits of the smartphone-based approach, namely the absence of a real head-tracking device, flexion-extension is considered only in combination with other movements. Game sequences last about 1 minute and at any time the subject can look at his current metric score without moving the head. Moreover, it is always possible to exit the game. In order to improve the immersivity experienced by a larger class of patients, a user is allowed to adapt different settings (ambient, music, etc...) for the same exercise. When a user starts CeVRicale-app, some safety warnings are provided and then the initial scene consists of a virtual physiotherapy center where it is possible to read the description of the games and to choose which one to play. At the end of an exercise, a user returns to the medical center, where his/her results are displayed and s/he can answer some questions concerning user satisfaction and perceived pain.

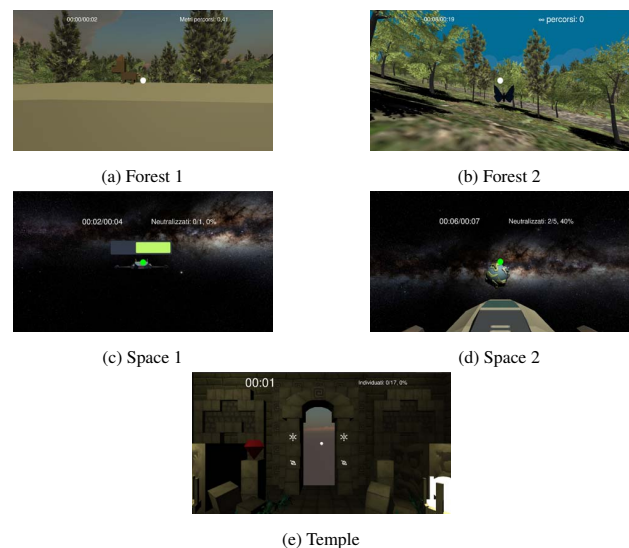


Figure 1: Screenshots from the five exergames.

2.1 Exergames

There is significant evidence of VR effectiveness in the treatment of chronic neck pain [3]. Our project provides more immersive settings than other exergame-based works (see [3]); this approach is promising but further investigation on the field with patients is required.

For the game "Forest 1", a user is asked to track a dog which moves horizontally. In order to reduce vertical oscillations, the animal moves only if it is being tracked. In this game, the metrics

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are: for how long a user was able to track the animal, and for how many meters the animals walked. This game focuses on rotation.

For the game "Forest 2", the user is asked to track a butterfly which flies in the shape of an ω . This is a standard movement in the cervical assessment field, see e.g. [8]. As before, the animal moves only if it is being tracked. In this game, the metrics are: for how long the animal was tracked and how many ω s the animal has done. This game focuses on rotation and inclination.

The two former games are addressed to a patient who looks for a relaxing moment in carrying out exercises. The two following ones are instead meant for patients looking for a higher degree of activity.

For the game "Space 1", the user is inside a spaceship and must point towards arriving enemies; when an enemy is pointed to, its ship blocks and every second it is hit by a laser. Every enemy has 5 life points, after which it is destroyed and another one is spawned. Enemy spaceships can spawn only along the horizontal axis. In this game, the metrics are: for how long a user pointed to an enemyship, and the percentage of destroyed enemies. The game focuses on rotation.

For the game "Space 2", a user flies a spaceship using rotation and inclination. The goal is to reach resources by flying upon them. Resources, to allow the exercise of pointing, are spawned in groups of five along the same direction. In this game, the metrics are: for how long a user pointed to a resource, and the percentage of acquired resources.

Finally, the game "Temple" is a pointing exercise. A user is located in a temple, together with 17 hidden gems, and must look and point at each of them. Gems appear progressively, starting from a small angle w.r.t the camera and increasing in difficulty. This is the only game that lasts for either 2:30 minutes or until all gems are located. In this game, the metrics are: the time to complete the exercise and the percentage of located gems.

In Table 1 we show an overview of games and their rehabilitative function.

Game	Metrics	Function
Forest 1	Tracking time, virtual meters	Rotation
Forest 2	Tracking time, virtual ω	Rotation, inclination
Space 1	Pointing time, % success	Rotation
Space 2	Pointing time, % success	Rotation, inclination
Temple	Overall time, % completion	All

Table 1: Games overview, with respective metrics and rehabilitative functions

2.2 Implementation details

CeVRicale is an Android app, developed using the Unity framework, for Google Cardboard. A demo version of the app, in Italian, is available for free ¹.

As in classic studies, such as [8], and virtual reality games, user's gaze is rendered as a virtual pointer whose position is fixed w.r.t the virtual camera. To prevent damage, we have designed non-intrusive mechanisms to assure safety of range and speed of movement. Every frame, i.e. every 15-20 ms depending on the device, if the angular speed of movement exceeds 5°, a sound signal reminds the user to slow down. Regarding the range of movement, all rotation exercises require a horizontal range of about 90°-120°; exercises including inclination require a vertical range of about 60°. Both of these

¹<https://site.unibo.it/varlab/en/student-demos/laboratorio-di-realta-virtuale-e-realta-aumentata>

measures are compliant with standard guidelines for a healthy patient [5].

3 DISCUSSION

Our project meets the requirements proposed by Faric et al [2]. The use of VR allows for intuitive controls, even for non-experienced users; games are immersive and involve more than one sense, allowing the patient to be distracted from PA. Indeed, we provide different settings in terms of different scenes and also soundtrack, which include different and diverse musical styles. Moreover, it has been shown (e.g. in [3]) that combining VR with rehabilitation has not only a positive effect on patient engagement, but it also extends the participation of the subject in the treatment. At the end of a game session, the user benefits from performing exercises deemed as equivalent to those experienced during a physiotherapy session. Moreover, safety is addressed both by having a short gaming duration (also to avoid motion sickness) and by implementing a feedback from the system in case dangerous movement patterns are detected. It is also possible for the patient to share their result and assessment with a practitioner in order to have constant monitoring of the health condition. CeVRicale is a proof-of-concept of supporting medium for cervical rehabilitation, combining VR with key principles of mobile-health (such as low price and ease of use).

4 CONCLUSION AND FUTURE WORK

This project is the first step of a study combining computer scientists, physiotherapists and patients to prove the effectiveness and engagement of the application. We suggest that the application may be helpful to tele-health, specifically for musculoskeletal rehabilitation of cervical spine. However, the project may be limited by the need to access a significant sample size and the few components of the group. We also look for further research on: implementing sociable aspects (such as the possibility of sharing results) and studying the applicability of our approach as a supporting device for the standard rehabilitation techniques for cervical spine, paying special attention to user satisfaction and application effectiveness.

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